

Order

Order

30 Points



Cipher

Cryptography

Today

#20

We intercepted one of Cipher's messages containing their next target. They encrypted their message using a repeating-key XOR cipher. However, they made a critical error—every message always starts with the header:

ORDER:

Can you help void decrypt the message and determine their next target?

Here is the message we intercepted:

1c1c01041963730f31352a3a386e24356b3d32392b6f6b0d323c22243f6373

1a0d0c302d3b2b1a292a3a38282c2f222d2a112d282c31202d2d2e24352e60

Easy Crypto

With the given clues we can say that the header is ORDER: and the hex message is given

First we will try to convert the header into hex and to binary and we will xor with the given message

4f 52 44 45 52 3a this is the hex of ORDER: now we will convert it into binary for XOR-ing.

```
00110100 01100110 00100000 00110101 00110010 00100000 00110100
00110100 00100000 00110100 00110101 00100000 00110101 00110010
00100000 00110011 01100001
```

This Is the binary of it similarly we will convert the given message to binary

```
00110001 01100011 00110001 01100011 00110000 00110001 00110000
00110100 00110001 00111001 00110110 00110011 00110111 00110011
00110000 01100110 00110011 00110001 00110011 00110101 00110010
01100001 00110011 01100001 00110011 00111000 00110110 01100101
00110010 00110100 00110011 00110101 00110110 01100010 00110011
01100100 00110011 00110010 00110011 00111001 00110010 01100010
00110110 01100110 00110110 01100010 00110000 01100100 00110011
00110010 00110011 01100011 00110010 00110010 00110010 00110100
00110011 01100110 00110110 00110011 00110111 00110011 00110001
01100001 00110000 01100100 00110000 01100011 00110011 00110000
00110010 01100100 00110011 01100010 00110010 01100010 00110001
01100001 00110010 00111001 00110010 01100001 00110011 01100001
```

00110011 00111000 00110010 00111000 00110010 01100011 00110010
01100110 00110010 00110010 00110010 01100100 00110010 01100001
00110001 00110001 00110010 01100100 00110010 00111000 00110010
01100011 00110011 00110001 00110010 00110000 00110010 01100100
00110010 01100100 00110010 01100101 00110010 00110100 00110011
00110101 00110010 01100101 00110110 00110000

This is the binary of message

The header length is 128-bits now we will XOR the key with first 128-bits of message.

I. Input: hexadecimal (base 16) ▾

1c1c01041963

II. Input: hexadecimal (base 16) ▾

4f524445523a

Calculate XOR

III. Output: ASCII (base 256) ▾

SNEAKY

So the word we got is **SNEAKY** this the key now we will XOR this key with the entire message as it is repeated XOR

I. Input: hexadecimal (base 16) v

```
1c1c01041963730f31352a3a386e24356b3d3
2392b6f6b0d323c22243f63731a0d0c302d3b
2b1a292a3a38282c2f222d2a112d282c31202
                                d2d2e24352e60
```

II. Input: hexadecimal (base 16) v

```
534e45414b59534e45414b59534e45414b595
34e45414b59534e45414b59534e45414b5953
4e45414b59534e45414b59534e45414b59534
                                e45414b59534e
```

Calculate XOR

III. Output: hexadecimal (base 16) v

```
4f524445523a2041747461636b20617420646
1776e2e205461726765743a2054484d7b7468
655f6861636b66696e6974795f68696768736
                                3686f6f6c7d2e
```

So we got the out put now we will decode it

|ORDER: Attack at dawn. Target: THM{the_hackfinity_highschool}.

We got the flag